Smaller Diophantine forms of Con(ZFC)

Lower degree, fewer unknowns, and smaller fixed coefficients

Research synthesis prepared for ProveIt

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Abstract

We combine the eight fixed-input packages in ProveIt's `con-zfc-diophantine-forms` collection with its ordinary strong-Pell, signed-quotient, and quadratic-root-elimination arguments. The resulting improvements are explicit arithmetic circuits: a four-equation system in 15 positive unknowns, of maximum degree 28 and 63 operations; its single degree-56 sum of squares, at 74 operations; single polynomials in 14 positive unknowns at 73 operations/degree 342 and 75 operations/degree 293; a quartic in 33 positive unknowns; and equations in 33 unrestricted integer unknowns. A separate finite-window reduction replaces nine-cell windows by projected crosses, reducing the exact coefficient-width parameter from 5^{67} to 5^{28}. Each completed construction has a solution exactly when the literal K215 machine halts on blank tape. The established syntactic calibration therefore gives, by mathematics formalizable in ZFC, that unsolvability is equivalent to Con(ZFC). This packet contains proofs, complete circuits, fixed coefficient definitions, and reproducible finite checks. It is not a new end-to-end Lean or Rocq verification, and it claims no global minimum.

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1 Results, comparison criteria, and scope

1.1 The theorem being constructed

Fix a conventional effective proof predicate for ZFC, with consistency meaning that no finite proof of falsity exists. All four coefficient ports in this report denote *fixed integers*, supplied by the terminating recipe in Section 6. There is no quantified program index, input, exponent, or computation predicate in the final equations.

Theorem 1.1 (Fixed Diophantine consistency forms)**.** *For each polynomial or system in Table 1, with its declared domain and the fixed coefficients of Section 6, ZFC proves*

$$\textit{the equation or system has no solution} \iff \mathrm{Con(ZFC)}.$$

The same arithmetic reductions also apply with the original valid inputless coefficients of package 06. The theorem does not assume Con(ZFC).

The proof has three components. The new algebraic reductions preserve existence of a solution. The new coefficient recipe encodes exactly the same blank K215 computation problem. The repository's K215 calibration identifies blank halting with a ZFC refutation. The last step is used as ordinary formalizable mathematics, not as an appeal to the soundness of Lean from inside ZFC; see Section 7.

Construction	Unknowns	Equations	Degree	Ops
Inherited P_{67}, new coefficients	15	1	155	67
Ordinary strong system	15	6	28	61
Grouped ordinary system	15	4	28	63
Ordinary product P_{69}	15	1	99	69
Four-square polynomial Q_{56}	15	1	56	74
Root-eliminated Ψ_{14}	14	1	342	73
Root-eliminated Φ_{14}	14	1	293	75
Root-eliminated system	14	5	244	69
Quadratic lift / its sum of squares	33	24 / 1	2 / 4	–
Unrestricted-integer $\widehat{P}_{67}$	33 $\mathbb{Z}$	1	292	130
Unrestricted-integer $\widehat{P}_{68}$	33 $\mathbb{Z}$	1	228	131

Table 1: Newly combined tradeoffs. Unmarked domains are strictly positive integers. Degree for a system means the maximum residual degree.

Every binary addition, subtraction, and multiplication is charged, including multiplication by a fixed numeral. Reading a fixed coefficient and reusing a register are free. In system counts, comparisons are free; their residual subtractions are charged when building a single polynomial. Fixed coefficients have degree zero. Circuit registers are polynomial abbreviations, not extra existential variables. Coefficient bit length is reported separately because the operation metric does not charge it.

For a compact system, the four-equation version is the clearest main recommendation. For one polynomial with fewer primary unknowns, use Ψ_{14} or Φ_{14}. The 67-operation evaluator remains the smallest operation count validated in this comparison, now with smaller coefficient definitions. No result below 67 operations was established. Classical nine-unknown existence results remain relevant: the repository's `Jones1982.nine_unknowns` proves an existence statement over the naturals, without a degree in its theorem statement [12, 13]. Thus fourteen is a local improvement over these delivered explicit circuits, not an absolute variable minimum.

1.2 What was already present

The source snapshot is pinned to commit

`3b201cdea32d2c816f5f3d20734051f25fd3ef14`.

The eight packages had themselves been developed against the earlier pin

`cdd582bec13f908df3711fa9e05d1ddac368a5db`.

The intake catalogue at the newer pin carefully distinguishes mathematical proofs from finite audits and proof-assistant builds [1]. We preserve those distinctions.

Package	Relevant contribution already present
01 explicit82	Fixed ordinary-input specialization; 18-unknown, 82-op polynomial.
02 system63	Ordinary strong Pell relation; 16 unknowns, seven equations, degree 28.
03 system64	Related 16-unknown system; root elimination yielding 15 unknowns.
04 poly73	Affine marker guard and a two-sided machine normalization.
05 fixed15–16	Explicit quadratic-norm root elimination; 15-unknown 80/82-op choices.
06 blank67	Raw K215, no ordinary input; 15 unknowns, six equations, 61 ops; polynomial 67.
07 fixed70	Signed factors and asymmetric scale choices; another low-degree system.
08 start-only67	Same 67-op arithmetic skeleton as 06; a 34-unknown quartic lift.

Table 2: The existing ideas combined here. Detailed original metrics and credits remain in the pinned catalogue.

Packages 06 and 08 already give the same 67-operation formula up to renaming, but use different coefficient integers. Packages 03 and 05 also share a 15-unknown root-elimination construction up to an affine slack change. We do not count either overlap as a new result. The changes here are the *combinations*: ordinary strong with inputless arithmetic; unit-sign grouping; inputless root elimination; a wider signed-coordinate domain; and a smaller window alphabet. Coefficients from different compilers must never be interchanged without proving the corresponding compiler theorem.

2 A common arithmetic interface

2.1 Fixed ports and primary unknowns

Write the four fixed integer ports as

$$\beta = B - 1, \qquad K = D_C + BD_R, \qquad M_C, \qquad M_F = M_F^{\mathrm{nat}} + B - 1.$$

In particular, M_F is the *shifted* source port. The recipe gives $B = 2^d$, $d \geq 4$, and

$$0 < M_C, M_F^{\mathrm{nat}} < B - 1, \qquad M_C \equiv 2 \pmod 4, \quad M_F^{\mathrm{nat}} \equiv 4 \pmod 8, \qquad (1)$$
$$\mathrm{pc}(M_C) + \mathrm{pc}(M_F^{\mathrm{nat}}) = d.$$

These numerical conditions are useful consequences of the full recipe; they do not by themselves characterize valid compiler coefficients.

Initially let

$$J, F, \alpha, z, f, h, i, T, s, w, \tau, \eta, \zeta, y, \gamma \in \mathbb{Z}_{>0}.$$

Here z is the transport quotient and T the auxiliary quotient. Define the following polynomial abbreviations:

$$\begin{aligned}
q &= \beta J + 1, & C &= q - F - \alpha, & & \\
X &= wq, & Y &= sq^3, & E &= XY, \\
k &= \eta + \zeta, & c &= kY + \eta, & a &= E + Y, \\
H &= 4a + 3, & \Delta &= a^2 + H = (a+1)(a+3), & & \\
D &= X + ac + \gamma H, & L &= E(kY), & & \\
G &= (q-1)(q-F) + \alpha = q(q-F) - C, & & & & \\
R &= G(q^2-1) + (M_C + qM_F)J, & & & & \\
V &= c(Tf - 1) - Rf^2. & & & &
\end{aligned} \tag{2}$$

Fixed exponents such as q^3 are ordinary polynomial operations. There are no variable exponents in any final circuit.

Four factors are common to all constructions:

$$\begin{aligned}
N_0 &= \tau^2 - L(L + k), & N_1 &= D^2 - \Delta c^2, \\
N_k &= k - hE - R, & N_t &= (K + w)C + q - F - z(q-1).
\end{aligned} \tag{3}$$

For the inherited normalized strong block set

$$S = \Delta i c^2, \quad Q = S^2, \quad N_a^{\mathrm{norm}} = Q(V^2 - y^2) + y^2, \quad N_s = f^2 - \Delta i^2 c^4. \tag{4}$$

The existing 67-operation polynomial is

$$P_{67} = N_0 N_1 N_a^{\mathrm{norm}} N_k N_t (\Delta f^2 - Q) - \Delta. \tag{5}$$

The identity

$$P_{67} = \Delta(N_0 N_1 N_a^{\mathrm{norm}} N_k N_t N_s - 1)$$

holds in the polynomial ring. On the positive domain $\Delta > 0$, so every normalized factor at a zero is an integer unit. The first, main, auxiliary, and strong norms cannot be -1; the index and transport signs are subsequently forced positive. The proof order matters: transport initially gives only $C \geq 0$. The boundary $C = 0, F = 2, z = 1, N_t = -1$ is not excluded until index recovery.

The inherited coefficient-reduced companion P_{68} replaces only the auxiliary coefficient Q by $\Delta(f^2 - 1)$, retaining Q in the scaled strong factor. It has degree 123. Every zero forces $N_s = 1$, so these coefficients agree at zeros. This is a zero-set argument, not a polynomial identity obtained by substituting an equation off its zero set.

2.2 A kernel fact used below

For an integer $A \geq 2$, write

$$\chi_A(n) + \psi_A(n)\sqrt{A^2 - 1} = (A + \sqrt{A^2 - 1})^n.$$

The repository's normalized and ordinary strong-rank lemmas, the half-binomial Pell ratio, and the native mask/convolution compiler are the inherited mathematical inputs [6, 7, 8, 4]. The new arguments below explain exactly where each is used. In particular, the signed-quotient proof is reproduced far enough to justify the root-eliminated polynomial's positive domain; no sign of a reconstructed root is assumed without proof.

3 Lower degree from the ordinary strong equation

The inputless arithmetic of packages 06 and 08 can be combined with the ordinary strong equation used in package 02 [2, 4, 5]. This improves degree while preserving the fifteen positive unknowns and the four fixed coefficient ports. Keep the previously defined polynomials $q, C, X, Y, E, k, c, a, H, \Delta, D, R, V$, and keep the first, main, index and transport factors N_0, N_1, N_k, N_t. Replace the strong and auxiliary block by the polynomial abbreviations

$$I = ic^2, \qquad W = \Delta(f^2 - 1), \qquad N_a = W(V^2 - y^2) + y^2. \tag{6}$$

Here I, W are computed expressions, not additional unknowns. In particular, the strong equation is $I^2 = W$, rather than the normalized equation involving $(i\Delta c^2)^2$.

Construction	Unknowns	Equations	Operations	Degree
Ordinary system	15	6	$34M + 27A = 61$	28
Grouped ordinary system	15	4	$36M + 27A = 63$	28
Four-square polynomial	15	1	$40M + 34A = 74$	56
Ordinary product	15	1	$39M + 30A = 69$	99
Quadratic lift	33	24	not optimized	2
Its sum of squares	33	1	not optimized	4

All unknown counts in this table use the strictly positive domain. The operation model is unchanged: each binary addition, subtraction or multiplication costs one, including multiplication by a numeral.

3.1 Six equations, four equations, and one polynomial

The ordinary six-equation system is

$$\boxed{N_0 = N_1 = N_a = N_k = N_t = 1, \qquad I^2 = W.} \tag{7}$$

Its literal circuit uses 61 operations, and its residual degrees, in the displayed order, are

$$22, \quad 18, \quad 28, \quad 7, \quad 2, \quad 22.$$

The auxiliary coefficient $W = \Delta(f^2 - 1)$ is intentionally used in (6). Replacing it off the zero set by its equal-on-solutions expression I^2 would increase the degree of the auxiliary residual.

Proposition 3.1 (Grouping the unit factors)**.** *Over positive integers, system* (7) *is tuplewise equivalent to the four equations*

$$\boxed{N_0 N_t = 1, \qquad N_1 N_k = 1, \qquad N_a = 1, \qquad I^2 = W.} \tag{8}$$

Proof. Each factor in either product equation is an integer unit. The first factor has the form $\tau^2 - U(U+1)k^2$, where $U = XY^2 > 1$; the consecutive-product negative Pell equation excludes the value -1. Also $\Delta = (a+2)^2 - 1$ is 0 or 3 (mod 4), so $N_1 = D^2 - \Delta c^2$ cannot be -1 (mod 4). Thus $N_0 = N_1 = 1$, and the product equations give $N_t = N_k = 1$. The converse is immediate. □

The grouped residuals have exact degrees $24, 25, 28, 22$. Therefore the particularly transparent single equation

$$\boxed{\begin{aligned} Q_{56} = {} & (N_0 N_t - 1)^2 + (N_1 N_k - 1)^2 \\ & + (N_a - 1)^2 + (I^2 - W)^2 = 0 \end{aligned}} \tag{9}$$

uses 74 operations and has degree 56. This is four operations fewer than the direct six-square finalizer, at the same degree and unknown count. The degree is exact uniformly in the permitted

fixed constants. Write $Q_0 = \beta J$ and $k_0 = \eta + \zeta$ for highest homogeneous forms. The unique degree-28 residual, $N_a - 1$, has highest form

$$w^2 k_0^2 s^4 Q_0^{14} T^2 f^4.$$

Consequently the degree-56 part of Q_{56} is $w^4 k_0^4 s^8 Q_0^{28} T^4 f^8$, which is nonzero since $\beta > 0$.

3.2 Why the ordinary strong equation is sufficient

This reduction requires a different proof order from a purely normalized strong-norm argument. One cannot simply divide the supplied i by Δ. The relevant ordinary strong-rank theorem is the one proved in the repository's relaxed auxiliary note [6] and used in package 02 [2].

First consider a positive solution of (7). Its exact transport equation forces $C > 0$. Indeed, if $C \le 0$, then

$$N_t = (K + w)C + 1 - F - (z - 1)(q - 1) \le 0,$$

contrary to $N_t = 1$. Thus $F + C = q - \alpha < q$. The native mask bounds and the same packing identity as in the inputless construction give

$$(2q - 1)(q^2 - 1) < R < q^4 - q^3, \qquad E \ge q^4 > R + q^3, \qquad a > R + 2. \tag{10}$$

Set $A = a + 2$ and $P_0 = 2XY^2 + 1$. The positive first and main norms supply indices $n, p \ge 1$ such that

$$k = 2\psi_{P_0}(n), \qquad c = \psi_A(p), \qquad D = \chi_A(p).$$

Since $P_0 > A$ and $c > k$, parameter monotonicity gives $p > n$. The first-index equation and $P_0 \equiv 1 \pmod{E}$ imply $2n \equiv R + 1 \pmod{E}$. Since $0 < R + 1 < E$, one has $n \ge (R + 1)/2$. Elementary Pell growth therefore proves, before any auxiliary argument,

$$c > A\Delta^2, \qquad c > 2p, \qquad c > 2R. \tag{11}$$

Now $I^2 = W$ implies

$$f^2 = 1 + \frac{i^2 c^4}{\Delta}, \qquad f > \frac{c^2}{A} > 2c.$$

Together with (10)–(11), this yields

$$W - (Rf^2 + c) = (\Delta - R)f^2 - \Delta - c > 0. \tag{12}$$

The auxiliary equation is $WV^2 - (W - 1)y^2 = 1$. If $|V| > 1$, then $V^2 - 1 = (W - 1)(y^2 - V^2)$ gives $|V| \ge 2W - 1$. But $V = c(Tf - 1) - Rf^2 > -(Rf^2 + c) > -W$, excluding negative such values. The values $0, 1, -1$ are excluded by the norm or by $V \equiv -c \pmod{f}$ and $2 < c < f/2$. Hence $V > 1$.

The relaxed strong-rank theorem now gives an index m with

$$f = \chi_A(m), \quad p \mid m, \quad c \mid m, \quad ic^2 = \Delta\psi_A(m), \quad m \ge c > 2p. \tag{13}$$

In particular $c^2 \mid f^2 - 1$, since $f^2 - 1 = ic^2\psi_A(m)$. The proof-only integers

$$o = cT - Rf, \qquad j = Tf - R\frac{f^2 - 1}{c} - 1$$

are integral and positive: $of = V + c > 0$ and $jc = V + R > 0$. They restore precisely $V = of - c = jc - R$.

The retained odd auxiliary-index and signed step-down argument therefore applies using (13); it needs $c \mid m$, not the stronger and unjustified assertion $c^2 \mid \psi_A(m)$. It gives $R \equiv \pm p \pmod{c}$. Since $0 < R, p < c/2$, this forces $p = R$. The no-wrap argument gives $2n = R + 1$. The lower ratio estimate and main projection then recover $X = 2^R$, followed by $q = 2^t$, the retained q^3-divisibility

of Y, and the half-binomial population threshold. The native masks, synchronization and forced Start now decode the finite accepting computation exactly as in the inputless proof [4, 5].

For completeness, take the canonical normalized solution of the inputless construction and replace only

$$i_{\text{ordinary}} = \Delta i_{\text{normalized}}.$$

Its strong equality becomes $I^2 = W$, and its auxiliary coefficient is unchanged. This is an existence-preserving map in the required direction; no division by Δ or coordinatewise equivalence of arbitrary strong-norm tuples is asserted.

3.3 A 69-operation product of degree 99

The ordinary unit factor is $N_s^{\mathrm{ord}} = 1 + I^2 - W$. The equation

$$\boxed{P_{69} = N_0 N_1 N_a N_k N_t N_s^{\mathrm{ord}} - 1 = 0} \tag{14}$$

has the same positive zero set as (7). It costs 69 operations and has degree 99. Its factor degrees are $22, 18, 28, 7, 2, 22$, and its nonzero leading form is

$$8\gamma h i^2 w^7 s^{16} k_0^9 Q_0^{55} T^2 f^4 (w(Q_0 - F - \alpha) - zQ_0).$$

Here the signed bootstrap must be stated separately. A zero makes every factor a unit. The first and main norm signs are positive as before. The factor N_s^{ord} cannot be $-1 \pmod 4$: when $\Delta \equiv 0$, it is $1 + I^2$, and when $\Delta \equiv 3$, it is $I^2 + f^2$. Thus $I^2 = W$, and the square coefficient in N_a also excludes its negative sign. Write $N_k = \varepsilon$, $N_t = \nu$, where $\varepsilon, \nu \in \{1, -1\}$ and $\varepsilon\nu = 1$.

Transport first gives only $C \geq 0$. Its exceptional zero case is $C = 0, F = 2, z = 1, \nu = -1$; retain it. The lower packing bound is still $(2q-1)(q^2-1) < R$: for $C > 0$, use $F + C \leq q - 1$, while in the exceptional case $q^2 - C - qF = q^2 - 2q > 2q - 1$. For the upper bound, the weaker estimate $q^2 - C - qF \leq q^2 - q$ suffices and gives

$$R < q^4 - q^3 + q^2 - 1, \qquad E \geq q^4 > R + q^3 - q^2 + 1 > R + 5.$$

Now $n \geq (R-1)/2$, $p > n$, and $k = R + \varepsilon + hE > 2R$ give (11). The ordinary gap, rank, auxiliary restoration and step-down proofs remain valid and give $p = R$. Writing $2n = R + \varepsilon + vE$, $v \geq 0$, a positive v would force $2n > 2R$, contradicting $n < p$. Thus $2n = R + \varepsilon$. If $\varepsilon = -1$, then $p = 2n + 1$; Pell duplication with $\chi_A(2) > P_0$ gives $c > Ak > k(Y + 1)$, contradicting the ratio slacks. Therefore $\varepsilon = \nu = 1$, eliminating the exceptional branch and proving all six equations. This proves the claimed equivalence without using $C > 0$ prematurely.

3.4 An explicit quartic in 33 positive unknowns

The ordinary strong equation also removes one register from the 34-variable quadratic lift of package 08 [5]. Besides the original fifteen positive variables, introduce eighteen positive register variables

$$N, E, P, c, a, X, D, C_2, G, L, R, F_2, V_0, V, g, I, B_a, \Delta.$$

In this subsection these letters are genuine unknowns. Only $q = \beta J + 1$, $k = \eta + \zeta$, $H = 4a + 3$, and $C = q - F - \alpha$ are affine abbreviations. Impose the following 24 quadratic equations:

$$\begin{aligned}
L &= q^2 - 1, & N &= q(L+1),\\
X &= wq, & a &= E + sN,\\
E &= X(a - E), & c &= k(a - E) + \eta,\\
P &= E(c - \eta) + k, & \tau^2 &= P(P - k) + 1,\\
D &= X + ac + \gamma H, & \Delta &= a^2 + H,\\
C_2 &= c^2, & D^2 - \Delta C_2 &= 1,\\
k &= R + 1 + hE, & G &= (q-1)(q-F) + \alpha,\\
R &= LG + (M_C + qM_F)J, & (K+w)C + q - F - z(q-1) &= 1,\\
F_2 &= f^2, & V_0 &= Tf - 1,\\
V &= cV_0 - RF_2, & g &= y^2 - V^2,\\
I &= iC_2, & B_a &= I^2,\\
B_a &= \Delta(F_2 - 1), & y^2 - B_a g &= 1.
\end{aligned} \tag{15}$$

Every residual has degree at most two, since the four compiler ports are fixed coefficients. If $e_1, \dots, e_{24}$ are these residuals, the single polynomial is

$$\boxed{Q_{33} = \sum_{j=1}^{24} e_j^2 = 0.} \tag{16}$$

Its degree is exactly four: the coefficient of τ^4 is 1. The artifact supplies every expanded residual and the expanded quartic, which has 224 formal terms before specializing its four coefficient ports.

The first equations recover $N = q^3$, $X = wq$, $a - E = sq^3$, $E = X(a - E)$, and the original main Pell data. The last equations give $B_a = I^2 = \Delta(f^2 - 1)$ and the ordinary auxiliary norm. Projection therefore yields (7). Conversely, each ordinary solution has the displayed register extension. All registers are positive; the nontrivial cases are $G > 0$, $V_0 > 0$, $V > 1$, and $g > 0$, the last following from $(W - 1)(y^2 - V^2) = V^2 - 1 > 0$. Thus positivity discards no solution.

The register saved relative to package 08 is its $K' = D^2 - 1 = \Delta C_2$. That lift used $I = iK'$, whereas the ordinary version uses $I = iC_2$. Deleting K' and replacing its two equations by $D^2 - \Delta C_2 = 1$ gives precisely the one-variable, one-equation reduction. Shifting all 33 coordinates by 1 gives the same degree-four equation over nonnegative integers. These reductions inherit the stated compiler and K215 calibration; they do not constitute a new formal proof-assistant build or a proof of consistency.

4 Removing the strong root: fourteen positive unknowns

Delete f from the fifteen-variable list, leaving

$$J, F, \alpha, z, h, i, T, s, w, \tau, \eta, \zeta, y, \gamma \in \mathbb{Z}_{>0}.$$

All definitions in (2) except V remain unchanged, as do the four factors (3). Introduce polynomial abbreviations

$$\begin{aligned}
u &= ic^2, & S &= \Delta u, & Q &= S^2, & \mathcal{D} &= 1 + uS,\\
v &= cT, & b_* &= c + R\mathcal{D}, & P_4 &= N_0 N_1 N_k N_t, &&\\
\mathcal{A} &= Q(\mathcal{D}v^2 + b_*^2 - y^2) + y^2 - 1, &&&&&&\\
\mathcal{B} &= 2Qvb_*, & \mathcal{K} &= \mathcal{A}^2 - \mathcal{D}\mathcal{B}^2. &&&&
\end{aligned} \tag{17}$$

The letter b_* here is a root-elimination expression, unrelated to the fixed radix width b in Section 6.

Theorem 4.1 (Two explicit fourteen-unknown equations)**.** *The following ordinary integer-coefficient polynomials have a positive zero exactly when K215 halts on blank tape:*

$$\boxed{\Phi_{14} = P_4(1+\mathcal{K}) - 1 = 0}, \tag{18}$$

$$\boxed{\Psi_{14} = [P_4(\mathcal{A}+1) - 1]^2 - \mathcal{D}(P_4\mathcal{B})^2 = 0}. \tag{19}$$

The supplied circuits use respectively $75 = 44M + 31A$ *and* $73 = 44M + 29A$ *operations, of exact degrees 293 and 342.*

This applies the quadratic norm of packages 03/05 to the smaller inputless base of 06/08 [3, 4, 5]. The formula for $\mathcal{B}$ specifies an integer polynomial; the circuit shares products and need not explicitly compute that abbreviation. For example,

$$\mathcal{D}\mathcal{B}^2 = 4(Q\mathcal{D}v^2)(Qb_*^2).$$

Soundness of Φ_{14}*.* At a positive zero, P_4 and $1+\mathcal{K}$ are integer units. Since $\mathcal{B}$ is even,

$$1 + \mathcal{K} \equiv 1 + \mathcal{A}^2 \pmod 4,$$

so this factor cannot be -1. Consequently $P_4 = 1$ and $\mathcal{A}^2 = \mathcal{D}\mathcal{B}^2$.

The denominator $\mathcal{B}$ is nonzero before any index decoding. Indeed $Q, c, T > 0$, and $\mathcal{D} = 1 + \Delta i^2 c^4 > c > 0$. If the integer expression $c + R\mathcal{D}$ vanished, $\mathcal{D}$ would divide c, which is impossible. Thus

$$f = \mathcal{A}/\mathcal{B} \in \mathbb{Q}, \qquad f^2 = \mathcal{D} \in \mathbb{Z}_{>0}.$$

A rational number with integral square is integral, by reducing its numerator and denominator to coprime terms. Thus f is a nonzero integer. With $V = vf - b_*$, direct expansion gives

$$f^2 - \Delta i^2 c^4 = 1,$$
$$QV^2 - (Q-1)y^2 = \mathcal{A} + 1 - \mathcal{B}f = 1.$$

These equalities and $P_4 = 1$ restore a zero of (5), provisionally allowing f to be signed. The signed-domain theorem of Section 5 proves $fT > 0$; because the retained T is positive, the restored f is positive. The inputless compiler theorem therefore supplies an actual blank halting computation. □

Soundness of Ψ_{14}*.* Expansion of (19) shows that its constant term 1 is divisible by P_4. Hence $P_4 = \pm 1$, and

$$f = \frac{P_4(\mathcal{A}+1) - 1}{P_4\mathcal{B}} = \frac{\mathcal{A} + 1 - P_4}{\mathcal{B}}$$

is a defined rational number whose square is $\mathcal{D}$. It is again a nonzero integer. The restored auxiliary norm is

$$QV^2 - (Q-1)y^2 = \mathcal{A} + 1 - \mathcal{B}f = P_4.$$

Since Q is a square, this norm is congruent modulo four either to y^2 or to V^2; it cannot be -1. Therefore $P_4 = 1$, and the same restoration and signed-domain proof apply. □

Conversely, at any positive zero of P_{67}, all six normalized factors are one. Thus $f^2 = \mathcal{D}$, $\mathcal{A} = \mathcal{B}f$, $\mathcal{K} = 0$, and $P_4 = 1$. Deleting f gives a zero of both new polynomials. The inverse is unique because $\mathcal{B} \neq 0$. In particular, the two new equations have identical positive zero sets in their fourteen retained coordinates. The useful all-ring identity

$$\Psi_{14} = P_4\Phi_{14} + (P_4 - 1)(2P_4\mathcal{A} - 1) \tag{20}$$

is included in the algebra audit, but its identity alone would not prove the zero-set equivalence without the integer-unit argument.

4.1 A five-equation form

The equivalent system

$$N_0 = N_1 = N_k = N_t = 1, \qquad \mathcal{K} = 0$$

has 14 positive unknowns, five equations, and a 69-operation shared evaluator. Its residual degrees are $22, 18, 7, 2, 244$. It is useful when variable or equation count is the objective; the four-equation ordinary system has a substantially lower degree.

4.2 Exact degree and coefficient certificates

Put $q_0 = \beta J$, $k_0 = \eta + \zeta$, and $t_0 = w(q_0 - F - \alpha) - zq_0$. Highest homogeneous parts give

$$\begin{aligned} (P_4)_{49} &= 8\gamma h k_0^3 w^5 s^8 q_0^{29} t_0, \\ (\mathcal{D})_{34} = i^2 k_0^4 w^2 s^6 q_0^{20}, &\qquad (b_*)_{38} = i^2 k_0^4 w^2 s^6 q_0^{23}(q_0 - F), \\ Q_{46} = i^2 k_0^4 w^4 s^8 q_0^{28}, &\qquad (\mathcal{A})_{122} = i^6 k_0^{12} w^8 s^{20} q_0^{74}(q_0 - F)^2. \end{aligned}$$

Here $\deg(\mathcal{D}\mathcal{B}^2) = 214 < 244 = \deg(\mathcal{A}^2)$. Therefore

$$\begin{aligned} (\Phi_{14})_{293} &= 8\gamma h i^{12} k_0^{27} w^{21} s^{48} q_0^{177}(q_0 - F)^4 t_0, \\ (\Psi_{14})_{342} &= 64\gamma^2 h^2 i^{12} k_0^{30} w^{26} s^{56} q_0^{206}(q_0 - F)^4 t_0^2. \end{aligned}$$

Both are nonzero for every $\beta > 0$. For example the coefficient of $J^{182}\gamma h i^{12}\eta^{27} w^{21} s^{48} z$ in Φ_{14} is $-8\beta^{182}$, and that of $J^{212}\gamma^2 h^2 i^{12}\eta^{30} w^{26} s^{56} z^2$ in Ψ_{14} is $64\beta^{212}$. These are uniform exact-degree certificates.

5 A larger signed domain and thirty-three integer unknowns

The root-elimination proof needs a domain theorem, not a choice of sign for a square root. The following statement also improves the independent unrestricted-integer construction.

Theorem 5.1 (Six coordinates may be signed)**.** *For either P_{67} or P_{68}, allow*

$$\tau, y, w, h, f, T \in \mathbb{Z}$$

while keeping $J, F, \alpha, z, i, s, \eta, \zeta, \gamma$ strictly positive. At every zero, $w, h > 0$, $\tau y f \neq 0$, and $fT > 0$. Replacing (τ, y, f, T) by $(|\tau|, |y|, |f|, \operatorname{sgn}(f)T)$ restores a positive zero.

We adapt the signed-quotient proof [9, 10] to the inputless arithmetic and add an elementary argument for signed h. The complete proof is also provided as `proofs/signed_domain.md`.

5.1 Automatic positivity before rank recovery

Because $q \geq B \geq 16$, $Y = sq^3 \geq q^3$, and $a = Y(wq + 1)$, integer w gives either $a \geq Y$ or $a \leq -Y(q - 1)$. Thus $\Delta = (a + 1)(a + 3) > 0$, even before w is known positive. The normalized product has integer-unit factors.

If $w \leq -1$, then $X, a, H < 0$ and $D = X + ac + \gamma H < ac < 0$. Hence $N_1 > -Hc^2 > 1$, contradicting its being a unit. If $w = 0$,

$$N_1 = H(2ac\gamma + \gamma^2 H - c^2), \qquad H = 4Y + 3 > 1,$$

again impossible for a unit. Consequently $w > 0$.

The negative first norm is excluded by the consecutive-product Pell descent. For the main and strong norms, $\Delta \equiv 0, 3 \pmod 4$ excludes -1; the auxiliary norm is congruent to a square modulo four. All four norms are therefore $+1$. This excludes $\tau, y, f = 0$. Apply the absolute-value and

simultaneous f, T replacement; only f^2 and Tf occur, so all polynomial values are unchanged. We can now take $f, \tau, y > 0$, keeping T signed. For P_{68}, the strong unit first gives $N_s = 1$, after which its auxiliary coefficient equals Q; the same argument applies.

Write $N_k = \epsilon$, $N_t = \nu$, where $\epsilon\nu = 1$. The transport expression is

$$N_t = (K + w)C + 1 - F - (z - 1)(q - 1).$$

Its being a unit forces $C \geq 0$. The actual mask bounds then give

$$3q + 3 < R < q^4 - q^3 + q^2 - 1 < q^4 - 2, \qquad E \geq q^4 > R + 2. \tag{21}$$

The index factor gives $k = hE + R + \epsilon$. A negative h makes $k < 0$, impossible. For $h = 0$, one would have $2 < k < q^4$. But the first Pell norm classifies

$$k = 2\psi_P(n), \qquad P = 2XY^2 + 1, \quad n \geq 1.$$

The value for $n = 1$ is 2, while $n \geq 2$ gives $k \geq 4P > q^4$. Hence $h > 0$.

5.2 Signed rank and the index signs

Put $A = a + 2$, so $\Delta = A^2 - 1$. The main norm gives $D = \chi_A(p)$, $c = \psi_A(p)$. Since $P > A$ and $c > k$, $p > n$. As $P \equiv 1 \pmod{E}$,

$$2n = R + \epsilon + v_0 E, \qquad v_0 \geq 0.$$

Thus $p > n \geq (R - 1)/2 \geq 24$, $c > 2p$, and the positive index quotient implies $c > 2R$.

The normalized strong norm gives $f = \chi_A(m)$, $\psi_A(m) = ic^2$. Strong divisibility of the Pell sequence implies $p \mid m$. Writing $m = pj$, its addition formula yields

$$\psi_A(pj)/c \equiv jD^{j-1} \pmod{c}.$$

The left side is divisible by c, and $\gcd(D, c) = 1$, so $c \mid j$. In particular $c \mid m$ and $m > 2p$.

The auxiliary norm classifies an odd index $\ell = 2v_1 + 1$ and a sign $e \in \{\pm 1\}$ such that

$$V = e\, C_{v_1}(S^2), \quad C_{v_1}(0) = (-1)^{v_1}\ell, \quad C_{v_1}(-\Delta) = (-1)^{v_1}\psi_A(\ell).$$

Here $C_j(s^2) = \chi_s(2j + 1)/s$ is an integer polynomial in s^2. The literal quotient expression, for either sign of T, gives

$$V \equiv -c \pmod{f}, \qquad V \equiv -R \pmod{c}.$$

As $S^2 \equiv -\Delta \pmod{f}$, squaring the first congruence and using the Pell doubling identity gives $\chi_A(2\ell) \equiv \chi_A(2p) \pmod{\chi_A(m)}$. The plus-sign step-down gives $2\ell \equiv \pm 2p \pmod{4m}$. One proves this by reducing the index modulo $2m$, reflecting it into $[0, m]$, and using $2\chi_A(m - 1) < \chi_A(m)$ to exclude a negative congruence between positive representatives. The endpoint m has residue zero and is also excluded. The second congruence, and $c \mid S, m$, now give $R \equiv \pm p \pmod{c}$. Since $R, p \in (0, c/2)$, $p = R$.

The no-wrap bound (21) forces $v_0 = 0$. If $\epsilon = -1$, then $p = 2n + 1$, whereas Pell duplication gives

$$\psi_A(2n) = 2A\psi_{2A^2-1}(n) \geq Ak,$$

because $2A^2 - 1 > P$. This contradicts $c = \psi_A(2n + 1) > Ak > k(Y + 1)$. Thus both index and transport units are $+1$; in particular $C > 0$, and R is odd.

5.3 Power recovery, masks, and the final sign

Write $r = (R-1)/2 \geq 24$ and $\xi = (X+1)^{2r}/X^r$. The elementary Pell estimates give $c/(k/2) > \xi$, hence $Y > X^r/3$ and $a > 2^R$. The projection recurrence gives $\chi_A(R) - a\psi_A(R) \equiv 2^R \pmod{H}$. The main-root definition therefore implies $X \equiv 2^R \pmod{H}$. Both positive quantities are below $a < H$, so $X = 2^R$. Consequently $q = 2^t = B^N$, and $q^3 \mid X$.

The upper ratio estimate uses only odd R, not a positive V:

$$0 < c/k - \xi/2 < 16r/(X+1) < 1/2.$$

Writing

$$\xi = M + \theta, \quad M = \binom{2r}{r} + \sum_{j=1}^{r} \binom{2r}{r+j} X^j, \quad 0 < \theta < 1/4,$$

one finds that M is even and $Y = M/2$. The exact binomial valuation gives $v_2(Y) = \mathrm{pc}(r) - 1$, so $q^3 \mid Y$ implies $\mathrm{pc}(R) \geq 3t + 2$.

Define the finite binary words

$$S_w = C - 1 + qF, \qquad T_w = (M_C J + 1) + q(M_F^{\mathrm{nat}} J - 1).$$

The packing identity is

$$R = (q^2 - S_w)(q^2 - 1) + T_w, \quad 0 < S_w < q^2, \quad \mathrm{pc}(T_w) = t + 2.$$

The inverse-population lemma gives $\mathrm{pc}(R) \leq 3t + 2$, with equality exactly when $S_w \,\&\, T_w = 0$. Thus $(C-1) \,\&\, (M_C J + 1) = 0$. Because $M_C J + 1 \equiv 3 \pmod 4$, this forces $C \equiv 1 \pmod 4$, and the literal packing identity gives $R \equiv 3 \pmod 4$.

The stronger step-down above has $\ell = \epsilon_0 R + 2mj_0$, $\epsilon_0 \in \{\pm 1\}$. Substitution into *both* unsquared auxiliary congruences gives

$$e(-1)^{r+mj_0} R \equiv -R \pmod{c},$$
$$e(-1)^{r+(m+1)j_0} c \equiv -c \pmod{f}.$$

As $c > 2R$ and $f > 2c$, both signs must equal -1. Their quotient forces j_0 even. Since r is odd, $e = 1$, so $V > 0$. Finally $cTf = V + c + Rf^2 > 0$ proves $T > 0$ in the transformed tuple, or $fT > 0$ before transformation. This proves Theorem 5.1 without applying a positive-domain compiler to a signed tuple.

5.4 Explicit conversion to unrestricted integers

Every positive integer is represented by

$$\mathrm{Pos}(u, v, t) = 1 + u^2 + v^2 + t(t+1), \qquad u, v, t \in \mathbb{Z}.$$

This is always positive. Surjectivity follows from the three-square theorem applied to $4n - 3$: its residue modulo four forces exactly one odd square, giving the displayed formula. Substitute a disjoint copy for each of the nine still-positive coordinates and keep the other six signed. The result has $9 \cdot 3 + 6 = 33$ unrestricted integer unknowns. Each substitution costs seven operations, so the counts are 130 and 131.

The exact degrees are smaller than naive degree doubling. Giving weight two to the nine substituted coordinates and weight one to the six signed ones, the P_{67} factor degrees are $42, 34, 112, 12, 4, 88$, totaling 292. For P_{68}, the auxiliary degree is 48, giving total 228. The nonzero top forms are respectively

$$8q_0^{92} h\gamma i^4 k_0^{13} w^{13} s^{26} T^2 f^2 z,$$
$$8q_0^{72} h\gamma i^2 k_0^{9} w^{11} s^{20} T^2 f^4 z.$$

Substitution of the disjoint positive quadratic forms preserves nonvanishing: setting the second and third new coordinates to zero gives the injective monomial substitution $x \mapsto u_x^2$.

6 Smaller coefficients from projected crosses

The coefficient compiler can discard information that the local computation predicate never reads. This gives a second, independent simplification of the arithmetic construction. Starting from the raw-machine blank tableau of package 06 [4], the full 3×3 window alphabet has 1 420 903 358 357 108 792 377 members. Five-cell crosses suffice, and the previous-time cell need only retain its horizontal-boundary bit. The resulting alphabet has exactly 87 923 161 members. The native polynomial still uses four fixed integer coefficients, while the exact binary length of $B - 1$ falls from 5^{67} to 5^{28}.

6.1 The local language and its projection

Use the same 439 cell symbols as package 06. Codes $0, 1$ are headless ordinary tape symbols. Code $2(q+1)+s$ has tape symbol $s \in \{0,1\}$ and head state $q \in \{0, \dots, 215\}$. The initialization symbols L, Q, R have codes $434, 435, 436$ and payloads $(0, \varnothing), (0,0), (0, \varnothing)$, respectively. The vertical and horizontal boundaries V, H have codes $437, 438$. The halt state 215 is padded by stationary transitions for tableaux only; the transition entering it retains its original write and move.

Let Loc be package 06's center predicate. It checks vertical-boundary propagation, separation of equal boundary types, horizontal-boundary propagation away from vertical boundaries, the initial phase path $L^u Q R^v$, the terminal halt condition, nonescape of the head, and the deterministic three-neighbor tape update. In its implementation `blank_helical.local_valid`, the four diagonal corners are unused. The previous-time neighbor is inspected only through the test that it equals H; its other data are unused.

Set $a_0 = 439$. A projected cross is

$$U = (p, \lambda, c, r, y) \in \{0,1\} \times \{0, \dots, a_0 - 1\}^4.$$

It is allowed when Loc accepts the 3×3 array with center c, left λ, right r, following-time center y, zero corners, and previous-time center H if $p = 1$ and 0 if $p = 0$. This is an exact finite predicate. Replacing the zero corners by any symbols, or replacing the previous-time representative by any symbol with the same H-indicator, leaves its truth value unchanged. Write $\eta(c) = 1$ if $c = H$ and 0 otherwise.

For a cross U, its right neighbor U', and its following-time neighbor U'', impose just

$$c = \lambda', \qquad r = c', \qquad \eta(c) = p'', \qquad y = c''. \tag{22}$$

Lemma 6.1 (Exact projected-cross reconstruction)**.** *For any cyclic length $N \geq 1$ and any integer stride h, projected crosses U_i satisfying (22) with neighbors U_{i-1} and U_{i-h} reconstruct a cell word c_i whose local computation predicate holds everywhere. Conversely every valid cell word gives such projected crosses. This remains true when offsets coincide, and requires neither $h \mid N$ nor coprimality.*

Proof. The four equalities and their shifts give, with subscripts modulo N,

$$\lambda_i = c_{i+1}, \quad r_i = c_{i-1}, \quad y_i = c_{i-h}, \quad p_i = \eta(c_{i+h}).$$

Thus U_i records exactly the data read by Loc at the cell c_i. Pull the word back to the plane by $(x,t) \mapsto -x - ht \pmod N$. The result is a valid plane with periods $(N, 0)$ and $(0, N)$. The converse follows by taking the indicated data from a cell word. These equalities are literal even for coinciding indices. □

The two distinguished tiles are

$$U_0 = (1, 434, 435, 436, 1), \qquad U_1 = (1, 0, 437, 438, 437).$$

Both are allowed by the center predicate. The first marks the initialized head: K215's first transition writes 1, moves right, and enters state 93, so the following-time center is the ordinary headless symbol 1. Every globally initialized Q has this projected cross. The second tile cannot occur globally. Its $p = 1$ forces its predecessor to be H, whereas its center is V. Vertical-boundary propagation at that predecessor would equate the vertical bits 0 and 1.

Order all allowed projected crosses as U_0, U_1, followed by the remaining five-tuples in ordinary lexicographic order. Package 06's helical padding proof now applies through the reconstruction lemma: there is a marked periodic word if and only if the raw K215 machine halts on blank input. Completeness permits independently chosen sufficiently large widths and heights, in particular powers of five. Soundness does not require the marked tile to be globally unique. The canonical completeness word has exactly one marked tile and no occurrence of the forbidden tile U_1.

6.2 The exact tile count

The inherited five-cell count is 38 256 607 897. Its 13 center classes were rechecked against the literal local predicate by the inherited finite census, considering every left/right pair for each class. Projection has constant fiber sizes. An ordinary interior center permits 438 non-H predecessors, all becoming one bit value. An initialization center permits only H, so its count is unchanged. At a boundary center the predecessor is unrestricted: 439 choices become two bit values. The resulting exact counts are:

Center class	Projected crosses
V	384 564
H	767 372
Headless 0	420 300
Halt with symbol 0	223 966
Headless 1	355 508
Halt with symbol 1	223 966
Working head, writes 0, moves left	11 570 268
Working head, writes 0, moves right	43 205 620
Working head, writes 1, moves left	5 046 428
Working head, writes 1, moves right	25 720 504
Initial L	2 118
Initial Q	437
Initial R	2 110
Total κ	87 923 161

This count includes both distinguished tiles. It performs no halting test and does not discard locally allowed tiles on the basis of global extendibility.

6.3 Six native payload slots

The native clause compiler copies the five coordinates of a tile and a sixth, derived coordinate $\eta(c)$. Its slots have the following exact exponent and clause-index maps. In every row s ranges from zero to one less than the slot size.

Slot	Size	Radix exponent $e(t, s)$	Clause exponent $j(t, s)$
Previous H-bit	2	$\kappa + 10a_0 + s$	$1 + s$
Left symbol	a_0	$\kappa + 5a_0 + s$	$3 + s$
Center symbol	a_0	$\kappa + 4a_0 + s$	$3 + a_0 + s$
Right symbol	a_0	$\kappa + 3a_0 + s$	$3 + 2a_0 + s$
Following symbol	a_0	$\kappa + a_0 + s$	$3 + 3a_0 + s$
Center H-bit	2	$\kappa + 7a_0 + s$	$3 + 4a_0 + s$

There are $4a_0+4$ payload positions. The sixth coordinate is fixed by the selected tile, so it is not an additional independent local choice. Denote the six values of U_i by $v_t(U_i)$, $0 \le t \le 5$.

Let $\ell \ge 2$ be least with $2^\ell > \kappa + 1$, and put $A_0 = 2^\ell$. Define

$$p_0 = \max\left(0, \left\lceil \frac{\kappa - 2\ell - 7a_0 - 10}{2} \right\rceil\right), \qquad j_{\max} = 4a_0 + 8 + p_0,$$
$$\mu = A_0 - 2 + \sum_{j=1}^{j_{\max}} A_0^j, \qquad m_0 = 2\ell + 11a_0 + 18 + 2p_0,$$
$$n_{\mathrm{pos}} = m_0 + 1, \qquad n_{\mathrm{dum}} = n_{\mathrm{pos}} - \kappa - 4a_0 - 8,$$
$$e_* = \kappa + 12a_0, \qquad M_0 = \kappa + 15a_0 + n_{\mathrm{dum}}.$$

Then $n_{\mathrm{dum}} \ge 1$. The allowed positions are the disjoint union

$$\mathcal{E} = \{0, \dots, \kappa - 1\} \cup \{e(t,s)\} \cup \{e_*, \dots, e_* + n_{\mathrm{dum}} - 1\} \cup \{M_0, 3M_0, 9M_0, 27M_0\}.$$

Their clause coefficients are

$$c_i = 1 + \sum_{t=0}^{5} A_0^{j(t, v_t(U_i))} + \sum_{u=1}^{4} A_0^{4a_0+4+u} \qquad (0 \le i < \kappa),$$
$$c_{e(t,s)} = A_0^{j(t,s)}, \qquad c_{3^{u-1}M_0} = A_0^{4a_0+4+u} \quad (1 \le u \le 4),$$

and dummy positions have coefficient zero. The bound

$$\sum_{e \in \mathcal{E}} c_e \le 11(\kappa + 4a_0 + 8) A_0^{4a_0+8}$$

follows by counting the constant, six payload terms and four anchors of each selector.

Set

$$E_{\max} = 27M_0, \qquad H_C = E_{\max} + 24M_0 + 3a_0 + 1,$$
$$T_1 = H_C + 2E_{\max} + a_0 + 1, \qquad T_2 = T_1 + 4E_{\max} + 2.$$

Let L_0 be the least power of five at least $T_2 + 2E_{\max} + 2$. For a positive integer x, write $\mathrm{bits}(x)$ for its binary length. Put

$$U_* = 4(n_{\mathrm{pos}} + 2)\big(22(\kappa + 4a_0 + 8) + 6\big) + 8,$$
$$b_{\min} = \max\{\ell(j_{\max} + 1) + 3, \ell(4a_0 + 8) + \mathrm{bits}(U_*) + 1, 4\}.$$

Let b be the least power of five at least $b_{\min}$, and set

$$\rho = 2^b, \qquad d = bL_0, \qquad B = \rho^{L_0} = 2^d.$$

6.4 The four fixed integers

Define

$$D_C = \rho^{3a_0} + \rho^{H_C + a_0} + \rho^{8M_0} + \rho^{24M_0} + \sum_{e \in \mathcal{E}} c_e(\rho^{T_1 - e} + \rho^{T_2 - e}),$$
$$D_R = \rho^{H_C},$$
$$F_0 = \mu(\rho^{T_1} + \rho^{T_2}) + \sum_{s=0}^{a_0-1} \rho^{\kappa + 4a_0 + s} + \sum_{s=0}^{1} \rho^{\kappa + 10a_0 + s} + \sum_{u=4}^{5} \sum_{s=0}^{a_0-1} \rho^{H_C + \kappa + ua_0 + s} + \rho^{9M_0} + \rho^{27M_0},$$
$$M_C = B - 1 - \sum_{\substack{e \in \mathcal{E} \\ e \ne 1}} \rho^e - 2\rho^{e_*}, \qquad F_{\mathrm{nat}} = F_0 + 4.$$

The polynomial's four coefficient ports are exactly

$$\boxed{\beta = B - 1, \qquad K = D_C + BD_R, \qquad M_C, \qquad M_F = F_{\mathrm{nat}} + B - 1.} \tag{23}$$

The last coefficient is the shifted mask required by the arithmetic core. All definitions are finite integer operations and bounded enumeration of an explicitly given predicate; there is no free machine index or oracle. These are four new integers and must be used together.

6.5 Why the native compiler proof survives

The four overlap tests in (22) are implemented by the same two fixed translations as before. Translation by $3a_0$ sends the following-symbol slot to the center-symbol slot, and the center H-bit slot to the previous H-bit slot. Translation by a_0 sends the right-symbol slot to the center-symbol slot and then to the left-symbol slot. The field masks test these four pairs; they contain $3a_0 + 2$ one-bit targets altogether. Two further targets test the anchor pairs.

The two clause bands enforce zero or one selector at each cell, its six matching one-hot payloads, and its four anchors. Indeed, the low A_0-digit test $A_0 - 2$, together with $A_0 > \kappa + 1$, permits only selector totals zero and one. Each higher one-bit clause requires its payload bit to equal the corresponding selected-tile bit. The added center H-coordinate is thereby forced to be $\eta(c)$. Parity of each sum of two Boolean payload bits is precisely equality, which supplies (22). Occupancy propagates along the horizontal neighbor relation from the marked origin to every cell.

There are no unintended convolution incidences. The six payload intervals are the explicit intervals in the preceding table; applying the two translations gives exactly the four pairs just listed. All other nonanchor positions lie strictly below M_0. At the anchor targets $9M_0, 27M_0$, the translations $8M_0, 24M_0$ see exactly $M_0, 3M_0$. The minimum clause exponent $T_1 - E_{\max}$ is larger than every copy or anchor test exponent. At a clause target, the only matching coefficient comes from the identical source position. The band separation prevents cross-contamination.

More explicitly, the geometry satisfies

$$T_2 - T_1 > 2E_{\max}, \qquad L_0 - (T_2 - T_1) > E_{\max}, \qquad T_2 + E_{\max} < L_0.$$

Thus an arbitrary rotation of native support, including the permitted upper-dummy spill at $e_* + 1$, cannot meet both clause bands. Among allowed support positions the difference $18M_0$ occurs only for the pair $(9M_0, 27M_0)$. There is no wrapped second possibility because $L_0 > 45M_0$. The inherited two-anchor argument therefore forces an arbitrary binary rotation first to radix alignment and then to whole-cell alignment. The exceptional half-radix case has only one odd spill and cannot satisfy the two anchor tests [8].

The radix bound gives

$$\rho \geq \max\left\{4(n_{\mathrm{pos}} + 2)\left(2\sum_e c_e + 6\right) + 8, 2\mu + 4, 16\right\}.$$

Hence the inherited carry and representative bounds apply, including the four-valued first dummy digit. Also

$$\mathrm{popcount}(\mu) = \ell + j_{\max} - 1, \qquad m_0 = 2\,\mathrm{popcount}(\mu) + 3a_0 + 4.$$

Consequently

$$0 < M_C, F_{\mathrm{nat}} < B - 1, \quad M_C \equiv 2 \pmod 4, \quad F_{\mathrm{nat}} \equiv 4 \pmod 8,$$
$$\mathrm{popcount}(M_C) = d - m_0 - 1, \quad \mathrm{popcount}(F_{\mathrm{nat}}) = m_0 + 1.$$

Their populations sum to d, exactly as required by the inverse-population step of the arithmetic proof. Selector 0 is permitted at the marked origin; selector 1 is forbidden. Its global impossibility was proved above, so this exclusion removes no canonical accepting tableau.

Finally, $T_2 - T_1 \equiv 2 \pmod 4$ and $\rho \equiv 2 \pmod 5$ make every paired clause contribution to D_C cancel modulo five. Because $a_0 \equiv 3 \pmod 4$,

$$2D_C - D_R \equiv 2(2 + 3\rho^{H_C} + 1 + 1) - \rho^{H_C} \equiv 3 \pmod 5.$$

Thus the inherited dummy-bit subset argument still tunes the actual Pell index to the intended time stride. The required width, height and d are powers of five. The lower exponents d change no logical step of that argument. The fixed low-field mask check also survives: the added payload slots all have positive exponents, so the unit radix digit still comes solely from the temporal neighbor's Start selector. The canonical word has exactly one such selector.

6.6 Numerical dimensions and scope of verification

The exact scalar parameters are

$$\begin{aligned}
\ell &= 27, & p_0 &= 43\,960\,012, & j_{\max} &= 43\,961\,776,\\
m_0 &= 87\,924\,925, & n_{\mathrm{dum}} &= 1, & e_* &= 87\,928\,429,\\
M_0 &= 87\,929\,747, & E_{\max} &= 2\,374\,103\,169, & & \\
H_C &= 4\,484\,418\,415, & T_1 &= 9\,232\,625\,193, & & \\
T_2 &= 18\,729\,037\,871, & b_{\min} &= 1\,186\,967\,982. & &
\end{aligned}$$

They give

$$b = 5^{13}, \quad L_0 = 5^{15}, \quad d = 5^{28} = 37\,252\,902\,984\,619\,140\,625.$$

The reduction in the binary length of $B - 1$ is the exact factor 5^{39}. The other ports satisfy $M_C < B$, $M_F < 2B$ and $K < B^2$; they too have binary lengths bounded by $2d$.

The supplied implementation evaluates these scalar parameters, derives the projected census from the checked 13-class count, checks the convolution incidences on small sparse fixtures, and proves the actual-dimension support incidences by exact integer interval intersections. In particular the support audit does not sample the 87 923 161 selectors: it reasons about their entire interval. It does not enumerate the full tile set, expand the four integers, produce an accepting K215 computation, or materialize a full Pell witness. The proof above is ordinary mathematics formalizable in ZFC. The existing K215 calibration then gives the requested equivalence; this section does not add a kernel-checked Lean or Rocq theorem or claim a proof of Con(ZFC).

7 Why ZFC proves the equivalence

The literal machine is the 215-state binary K215 table whose SHA-256 is

`fdbcef0825886035c3091c8537a69535bc96f9e265f0ae45863543d40ae70090`.

The inspected endpoint has the statements

```
KEndpoint215.blank_halts_iff_zfc_refutation
KEndpoint215.nonhalting_iff_zfc_consistency.
```

They use the syntactic predicate `BProv ZFCax_s [] Form.fBot`. The machine-specific proof invokes the common K endpoint and the constructible-choice equivalence between finite ZF and ZFC refutations; it is not an assertion that consistency is true [11].

The resulting ordinary mathematical chain is

$$\begin{aligned}
\exists \text{ Diophantine solution} &\iff \exists \text{ finite accepting local tableau}\\
&\iff \text{K215 halts on blank tape}\\
&\iff \exists p\ \mathrm{Proof}_{\mathrm{ZFC}}(p, \bot).
\end{aligned}$$

Negating this equivalence yields the theorem in the requested direction. The finite table, coefficient algorithms, polynomial manipulations, integer divisibility arguments, Pell recurrences, and finite-computation calibration can all be represented in ZFC. The mathematical argument does not require ZFC to prove that Lean or Rocq is sound. Nor does it infer a ZFC-internal proof merely from an external proof-assistant theorem: the intended internal proof is the ordinary syntactic calibration together with the explicitly described arithmetical reduction.

Formal status. This packet provides an ordinary proof using the pinned compiler/Pell development and the new lemmas proved here. It does not contain a new machine-checked term for the complete Diophantine theorem, or an encoded derivation in ZFC's own proof syntax. Lean and Rocq endpoint source and historical verification records were inspected; their full builds were not rerun. The native compiler itself is a mathematical research development, not a completed Lean compiler certificate. These are material verification boundaries, shared with the eight predecessor packages.

Incompleteness. If ZFC is consistent, the displayed equation really has no solution, but ZFC cannot prove that unsolvability statement without proving its own consistency. Mere consistency does not justify a claim that ZFC also cannot prove the opposite existential assertion; a corresponding soundness assumption is needed for that stronger two-sided independence claim. No consistency assumption is used in constructing the equivalence.

8 Reproducibility and what the checks establish

The archive contains the literal K215 table, the four-port coefficient specification, every arithmetic instruction, the quadratic residuals and expanded quartic, proof notes, inherited evidence, and provenance hashes. Run

```
python3 code/verify.py
```

to reproduce the bounded checks with Python's standard library. Rebuild this report with `make pdf`. The report's TeX and all section files are included. No network access is needed by verification. The default run checks and reuses the delivered exact local-predicate census; use `python3 code/verify.py --full` to rerun all thirteen center classes and their left/right pairs.

The checks cover the following independently auditable claims:

1. K215's exact table hash and the finite local-predicate census; projected alphabet counts by exact fiber sizes, rather than enumeration of all projected tiles.
2. Actual-dimension interval support incidences, clause-band separation, anchor differences, mask population formulas, and the fixed modulo-five unit; smaller sparse examples exercise the same coefficient construction.
3. Acyclic source binding, every live gate and port, exact operation counts, and exact source/formula identities on signed rational or integer diagnostic assignments.
4. Exact sparse expansion of the 24 quadratic equations and their quartic, including the coefficient 1 on τ^4.
5. Exact polynomial evaluation on generic lines over a finite field, corroborating degrees whose uniform upper bounds and nonzero leading forms are proved in the text.

None of these finite checks is a test of consistency, a proof by sampling, or an exhibited K215 halting computation. The enormous coefficient integers are not materialized. The output is fully specified by finite expressions and algorithms; it is not a practical expanded decimal polynomial.

8.1 Recommended incorporation

Retain the four-equation system and Q_{56} as the readable low-degree entry point, Ψ_{14} and Φ_{14} as the fourteen-variable alternatives, and the projected-cross recipe as an independent coefficient improvement. Keep the 67-operation circuit as a baseline. The source comments and proof notes distinguish the normalized strong coordinate from the ordinary strong coordinate: the conversion is an existence equivalence, and arbitrary ordinary solutions must not be divided by Δ without the relaxed-rank proof.

The next consequential formalization task is the common inputless compiler interface plus the signed-domain theorem. That would certify several of these tradeoffs at once. A search for fewer operations should work from the complete four-port circuit and its domain proof; deleting a Pell or population condition merely because it holds on a few examples does not establish a valid reduction.

References

[1] ProveIt contributors. *Con(ZFC) in Diophantine form*, eight-package catalogue, pinned 22 October 2026. Repository source.

[2] ProveIt contributors. *A fixed Diophantine representation of Con(ZFC)*, package 02-system63, report and ordinary strong proofs, same pin.

[3] ProveIt contributors. *Fixed-input Diophantine forms of ZFC consistency*, package 05-fixed15-16, Sections 4–7 and emitted circuits, same pin.

[4] ProveIt contributors. *Explicit Diophantine equations for Con(ZFC)*, package 06-blank67, report, fixed-blank proof, helical proof, and coefficient compiler, same pin.

[5] ProveIt contributors. *Explicit Diophantine equations for the consistency of ZFC*, package 08-start-only67, including `start_only_quartic34_proof.md`, same pin.

[6] ProveIt contributors. *Pell relaxed auxiliary proof*, `Computability/HilbertTenthProblem/Papers/1980/PELL_RELAXED_AUXILIARY_PROOF.md`, same pin.

[7] ProveIt contributors. *An auxiliary quotient gives a complete universal polynomial in 85 operations*, `complete85_auxiliary_bezout_projection.md`, native-stream-queue research notes, same pin.

[8] ProveIt contributors. *Modified native masks for the half-binomial complete75 compiler*, `complete75_half_binomial_compiler.md`, same pin.

[9] ProveIt contributors. *Signed auxiliary quotient absorption*, `complete84_signed_quotient_absorption.md`, especially Section 2, same pin.

[10] ProveIt contributors. *Signed auxiliary quotient: source-specific recovery of the positive domain*, `complete84_signed_quotient_soundness_scout.md`, and its independent review, same pin.

[11] ProveIt contributors. K215 endpoint and verification records. KEndpoint215.lean; verification directory.

[12] ProveIt contributors. `Jones1982.nine_unknowns`, `Computability/HilbertTenthProblem/Lean/Diophantine/Paper1982/NineMain.lean`, same pin.

[13] J. P. Jones. Universal Diophantine equation. *Journal of Symbolic Logic* 47(3), 549–571 (1982). doi:10.2307/2273588.